

Delhi Public School

Assessment Paper

Name: _____ Date: _____ Class: _____

Assignment Questions

For each question, read the text carefully and select the best answer from the given options. Each question carries 1 mark.

- Q1.** What is the main objective of the course-based project titled "AIRCRAFT TYPE IDENTIFICATION FROM SAR SATELLITE RADAR IMAGERY"? A) To design a new SAR satellite. B) To develop an air traffic control system. C) To identify different aircraft types from SAR satellite radar imagery. D) To create a comprehensive database of aircraft specifications. [1 M]
- Q2.** According to the provided material, who is the Head of Department (HoD) for CSE (AIML & IoT), R & AI at VNR VJIEET, Hyderabad? A) DR. Chalumuru Suresh. B) Dr. Sagar Yeruva. C) A. Vaishnavi. D) B. Sangeetha. [1 M]
- Q3.** Which of the following is NOT explicitly listed as a keyword for the project in the ABSTRACT section? A) Synthetic Aperture Radar (SAR). B) Aircraft Classification. C) Machine Learning. D) Deep Learning. [1 M]
- Q4.** What is identified as a primary challenge in aircraft classification from SAR images, as stated in the Problem Statement? A) The high computational cost of SAR image processing. B) The lack of real-time SAR satellite data acquisition. C) SAR images being inherently noisy and lacking rich visual features. D) The difficulty in obtaining human expert annotations for SAR images. [1 M]
- Q5.** Which specific preprocessing techniques were developed in the project to reduce speckle noise in SAR images? A) Histogram equalization and Gaussian blurring. B) Log transformation and Lee filtering. C) Median filtering and Wiener filtering. D) Contrast stretching and Adaptive thresholding. [1 M]
- Q6.** Based on the 'Results Obtained' section, which deep learning model achieved the highest test accuracy for aircraft classification? A) VGG16. B) EfficientNet-B0. C) ResNet-18. D) MobileNetV2. [1 M]
- Q7.** What technique was integrated into the system to provide model interpretability by generating heatmaps that highlight important regions in the input image? A) Principal Component Analysis (PCA). B) t-Distributed Stochastic Neighbor Embedding (t-SNE). C) Gradient-weighted Class Activation Mapping (Grad-CAM). D) Local Interpretable Model-agnostic Explanations (LIME). [1 M]
- Q8.** Which of the following is NOT explicitly listed as a limitation of the project in the 'Limitations' section? A) The dataset size is limited. B) Misclassification occurs between structurally similar aircraft types. C) The system is unable to process images from new, unrepresented aircraft types. D) The system is restricted to predefined aircraft classes. [1 M]
- Q9.** The project was carried out under the guidance and supervision at which specific institute, affiliated with Jawaharlal Nehru Technological University Hyderabad (JNTUH)? A) Indian Institute of Technology Hyderabad. B) Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology. C) National Institute of Technology Warangal. D) University College of Engineering, JNTUH. [1 M]
- Q10.** For which specific course was this project submitted in partial fulfillment of the requirements for the award of marks? A) Data Structures and Algorithms. B) Computer Networks. C) Neural Networks and Deep Learning. D) Operating Systems. [1 M]